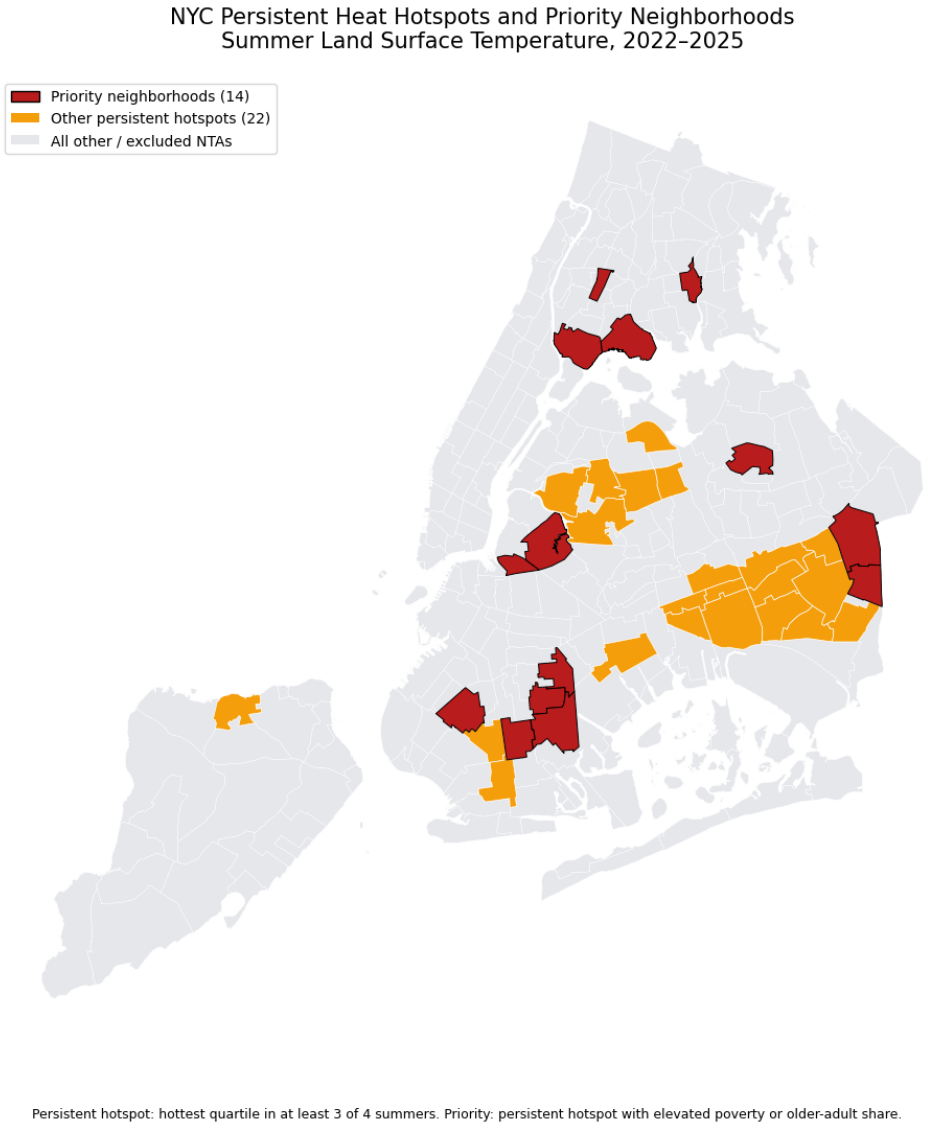
NYC Urban Heat & Environmental Vulnerability

Geospatial analytics portfolio case study | 2022–2025

THE CHALLENGE

Identify where persistently high summer surface temperatures overlap with selected socioeconomic vulnerability indicators, and translate satellite observations into a transparent neighborhood-level planning screen.

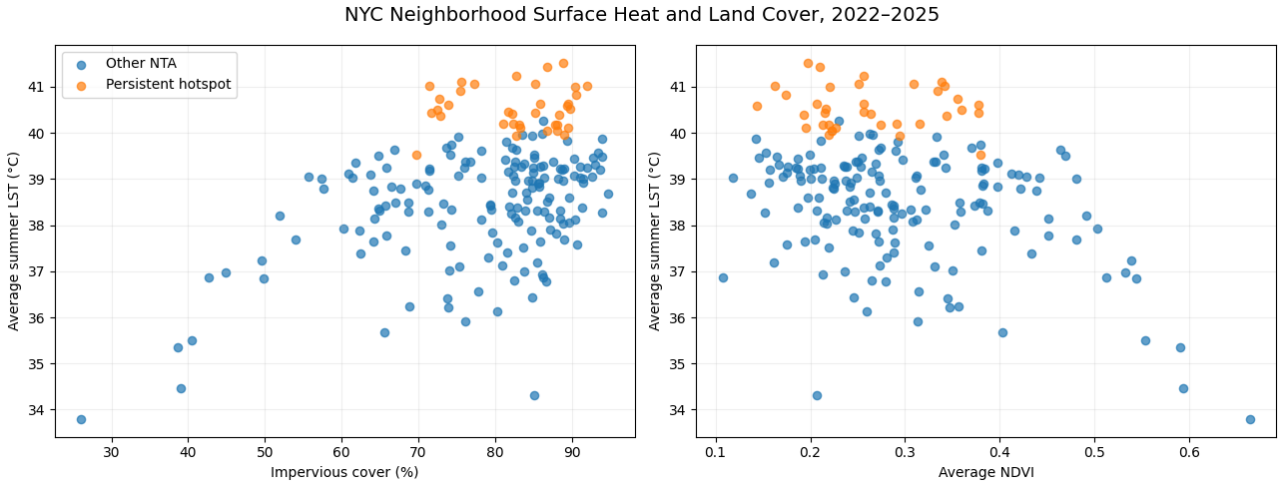
197 NTAs	36 hotspots	14 priorities
Complete analytical sample	Hottest quartile in ≥3 summers	Hotspots + elevated poverty or age 65+



NYC screening map: persistent hotspots (orange) and priority neighborhoods (red).

Approach and evidence

Workflow: Process Landsat 8/9 thermal imagery in Google Earth Engine; compute month-balanced June–August LST for 2022–2025; aggregate to NYC NTAs; join Landsat NDVI, NLCD imperviousness, and ACS demographic indicators; flag persistent heat and social-vulnerability overlap; test an additional permanent-water mask.



Spearman correlations: impervious cover +0.283, NDVI −0.238, poverty +0.182, older-adult share −0.171. Associations are descriptive and modest.

Results

Persistent hotspots averaged 40.54°C summer surface temperature versus 38.38°C in other analyzed neighborhoods, with 5.15 percentage points more impervious cover and lower average NDVI. All 36 hotspots and 14 priorities retained their persistence classification after revised permanent-water masking (mean absolute LST change ≈0.03°C annually).

Planning value and caveats

Provides a reproducible starting point for shade audits, greening feasibility, and neighborhood-specific resilience assessments. Surface temperature is not air temperature; neighborhood averages do not measure individual exposure or establish causal intervention benefits.

TOOLS Python · pandas · GeoPandas · Google Earth Engine · Landsat · GIS · Matplotlib